Assignment 3: Question 3, part 1

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt

# Dataset
X = np.array([600, 800, 1000, 1200, 1400, 1600])
y = np.array([45, 58, 72, 85, 96, 110])

# Add column of ones for intercept
X_b = np.c_[np.ones((len(X), 1)), X]

# Normal Equation
theta = np.linalg.inv(X_b.T @ X_b) @ X_b.T @ y

intercept = theta[0]
slope = theta[1]

print("Intercept:", intercept)
print("Slope:", slope)

# Prediction
y_pred = X_b @ theta

# Predict for 1500 sq.ft.
house = np.array([1, 1500])
prediction = house @ theta

print("Predicted Price for 1500 sq.ft:", prediction)

# MSE
mse = np.mean((y - y_pred) ** 2)
print("MSE:", mse)

# R2 Score
ss_total = np.sum((y - np.mean(y)) ** 2)
ss_residual = np.sum((y - y_pred) ** 2)

r2 = 1 - (ss_residual / ss_total)
print("R2 Score:", r2)

# Plot
plt.figure(figsize=(8,5))
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, y_pred, color='red', linewidth=2, label='Regression Line')

plt.xlabel("House Size (sq.ft)")
plt.ylabel("Price (Lakhs)")
plt.title("Linear Regression using Normal Equation")
plt.legend()
plt.grid(True)

plt.show()
```

```
Intercept: 6.638095238095263
Slope: 0.06457142857142857
Predicted Price for 1500 sq.ft: 103.49523809523812
MSE: 0.45079365079365136
R2 Score: 0.9990741344461107
```